Chapter 11: Project Risk Management

Information Technology Project Management, Ninth Edition

Note: See the text itself for full citations

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Learning Objectives (1 of 2)

- Explain the concept of risk as it relates to project management, and list the advantages of managing project risks according to best practices
- Discuss the elements of planning risk management and the contents of a risk management plan
- List common sources of risks on information technology (IT) projects
- Describe the process of identifying risks and create a risk register and risk report
- Discuss qualitative risk analysis and explain how to create probability/impact matrixes, and apply the Top Ten Risk Item Tracking technique to rank risks

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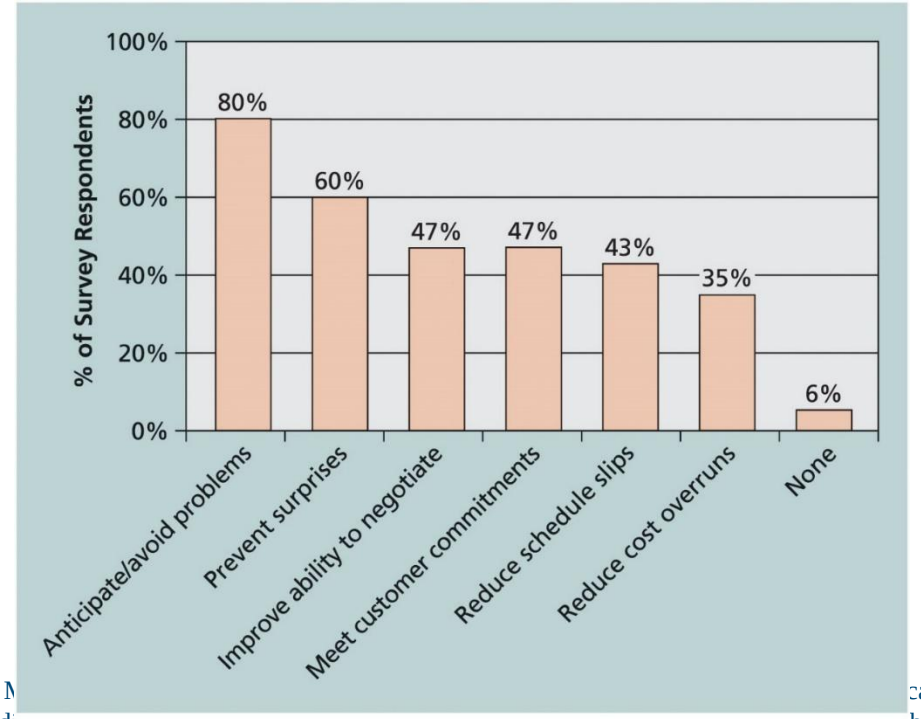
Learning Objectives (2 of 2)

- Explain quantitative risk analysis and how to apply decision trees, simulation, and sensitivity analysis to quantify risks
- Provide examples of using different risk response planning strategies to address both negative and positive risks
- Discuss how to monitor risks
- Describe how software can assist in project risk management
- Discuss considerations for agile/adaptive environments

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The Importance of Project Risk Management

- Project risk management is the art and science of identifying, analyzing, and responding to risk throughout the life of a project and in the best interests of meeting project objectives
  - Risk management is often overlooked in projects, but it can help improve project success by helping select good projects, determining project scope, and developing realistic estimates



Source: Kulik and Weber, KLCI Research Group

FIGURE 11-1 Benefits from software risk management practices

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Risk

- A dictionary definition of risk is "the possibility of loss or injury"
- General definition of a project risk is "an uncertainty that can have a negative or positive effect on meeting project objectives"
- Negative risk involves understanding potential problems that might occur in the project and how they might impede project success
  - E.g. completing work later or more costly than planned
  - Negative risk management involves possible actions to take to avoid, lessen, change, or accept the potential effects of risks on the projects
- Positive risks are risks that result in good things happening; sometimes called opportunities
  - E.g. completing work sooner or cheaper than planned
  - Positive risk management is like investing in opportunities

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Risk Utility

- Amount of satisfaction or pleasure received from a potential payoff
  - Utility rises at a decreasing rate for people who are risk-averse, having a lower tolerance for risk
  - Those who are risk-seeking have a higher tolerance for risk and their satisfaction increases when more payoff is at stake
  - Those who are risk-neutral achieve a balance between risk and payoff, doing a series of analyses when making decisions

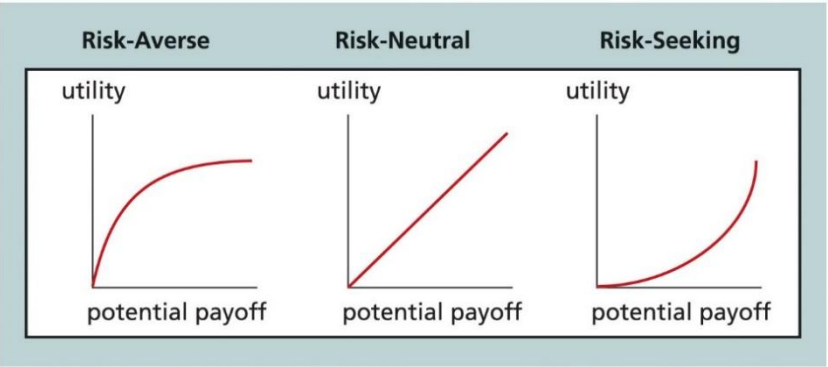
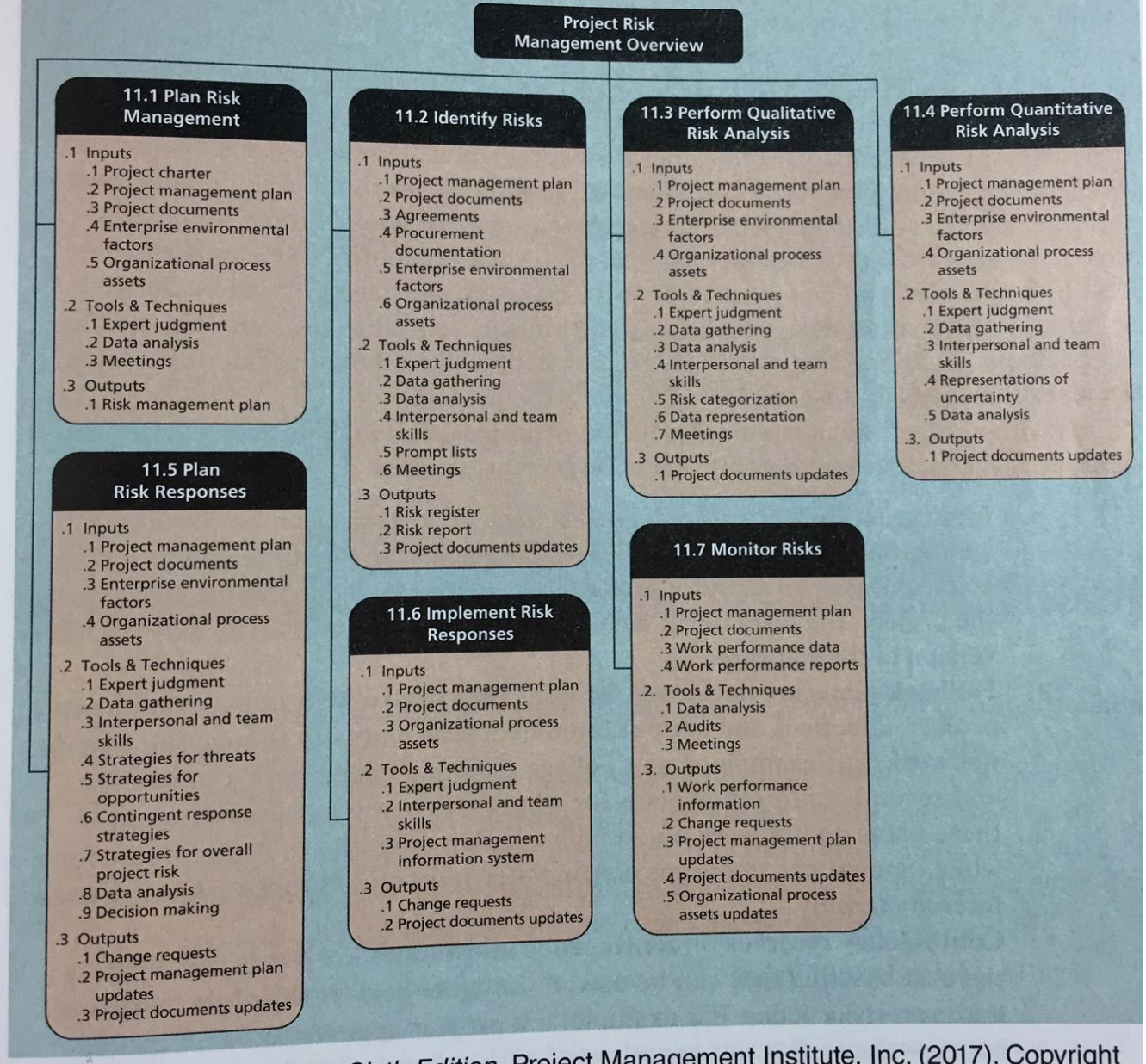


FIGURE 11-2 Risk utility function and risk preference

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Project Risk Management Processes

1. Planning risk management
   - Deciding how to approach and plan the risk management activities for the project
2. Identifying risks
   - Determining which risks are likely to affect a project and documenting the characteristics of each
3. Performing qualitative risk analysis
   - Prioritizing risks based on their probability and impact of occurrence
4. Performing quantitative risk analysis
   - Numerically estimating the effects of risks on project objectives
5. Planning risk responses
   - Taking steps to enhance opportunities and reduce threats to meeting project objectives
6. Implementing risk responses
   - Implementing the risk response plans
7. Monitoring risk
   - Monitoring identified and residual risks, identifying new risks, carrying out risk response plans, and evaluating the effectiveness of risk strategies throughout the life of the project



Source: *PMBOK® Guide – Sixth Edition*. Project Management Institute, Inc. (2017). Copyright and all rights reserved. Material from this publication has been reproduced with permission of PMI.

FIGURE 11-3 Project risk management overview

1. Planning Risk Management (1 of 2)

- Main output of this process is a risk management plan
  - Documents the procedures for managing risk throughout a project
- The project team should review project documents as well as corporate risk management policies, risk categories, lessons-learned reports from past projects, and templates for creating a risk management plan
  - It is also important to review the risk tolerances of various stakeholders

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Planning Risk Management (2 of 2)

Topic	Questions to Answer
Methodology	How will risk management be performed on this project? What tools and data sources are available and applicable?
Roles and responsibilities	Which people are responsible for implementing specific tasks and providing deliverables related to risk management?
Budget and schedule	What are the estimated costs and schedules for performing risk-related activities?
Risk categories	What are the main categories of risks that should be addressed on this project? Is there a risk breakdown structure for the project? (See the information on risk breakdown structures later in this chapter.)
Risk probability and impact	How will the probabilities and impacts of risk items be assessed? What scoring and interpretation methods will be used for the qualitative and quantitative analysis of risks? How will the probability and impact matrix be developed?
Revised stakeholders' tolerances	Have stakeholders' tolerances for risk changed? How will those changes affect the project?
Tracking	How will the team track risk management activities? How will lessons learned be documented and shared? How will risk management processes be audited?
Risk documentation	What reporting formats and processes will be used for risk management activities?

Table 11-2 Topics addressed in a risk management plan

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2. Identifying Risks

- Understanding what potential events might hurt or enhance a particular project
  - You cannot manage risks if you do not identify them first
  - It is important to identify potential risks early, and also continue to identify risks based on changing project environment
- Output
  - Risk register
  - Risk report

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Common Sources of Risk on IT Projects (1 of 3)

- Several studies show that IT projects share some common sources of risk
  - Market risk
    - E.g. Will the new product or service be useful or marketable?
  - Financial risk
    - E.g. Will the project meet the NPV, ROI, and payback estimates?
  - Technology risk
    - E.g. Will the project use leading-edge or bleeding-edge technology?
    - Can be broken down into hardware, software, and network risk
  - People risk
    - E.g. Does the organization have people with appropriate skills to complete the project successfully?
  - Structure/process risk
    - E.g. What degree of change will the new project introduce into user areas and business procedures?
- Useful to review the WBS to see if there might be specific risks by WBS categories
  - Identifying risks based on the nature of the project and product as well as project management knowledge areas (e.g. scope, time, cost, quality)
- May review risks in previous projects

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Common Sources of Risk on IT Projects (2 of 3)

- Risk breakdown structure is a hierarchy of potential risk categories for a project

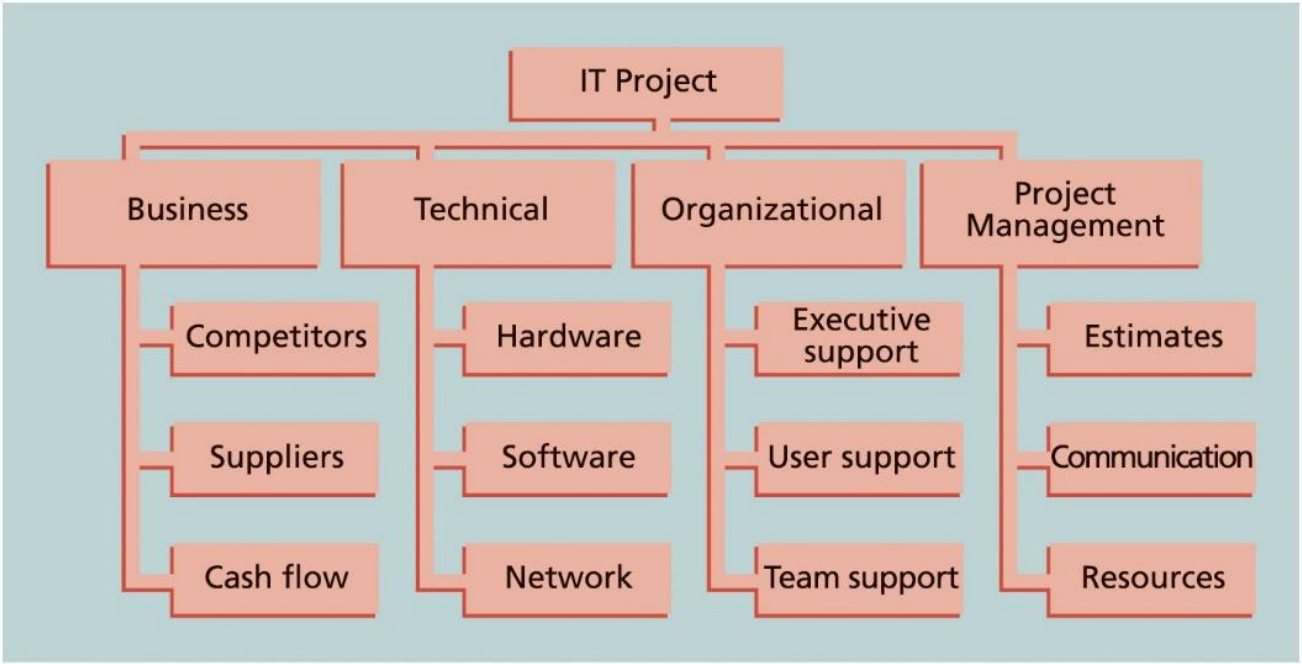


FIGURE 11-4 Sample risk breakdown structure

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Common Sources of Risk on IT Projects (3 of 3)

Knowledge Area	Risk Conditions
Integration	Inadequate planning; poor resource allocation; poor integration management; lack of post-project review
Scope	Poor definition of scope or work packages; incomplete definition
Time	Errors in estimating time or resource availability; errors in determining the critical path; poor allocation and management of slack; early release of competitive products
Cost	Estimating errors; inadequate productivity, cost, change, or contingency
Quality	Poor attitude toward quality; substandard design, materials, and workmanship; inadequate quality assurance program
Human resource	Poor conflict management; poor project organization and definition of responsibilities; absence of leadership
Communications	Carelessness in planning or communicating
Risk	Ignoring risk; unclear analysis of risk; poor insurance management
Procurement	Unenforceable conditions or contract clauses; adversarial relations
Stakeholders	Lack of consultation with key stakeholder; poor sponsor engagement

Table 11-3 Potential negative risk conditions associated with each knowledge area. *Source: R.M. Wideman

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Tools and Techniques for Identifying Risks (1 of 3)

- Brainstorming
  - Group attempts to generate ideas or find a solution for a specific problem by collecting ideas spontaneously and without judgment
  - An experienced facilitator should run the brainstorming session
  - Be careful not to overuse or misuse brainstorming
    - Psychology literature shows that individuals produce a greater number of ideas working alone than they do through brainstorming in small, face-to-face groups
    - Group effects (e.g. fear of disapproval, domination of other people) often inhibit idea generation

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Tools and Techniques for Identifying Risks (2 of 3)

- Delphi technique
  - Used to derive a consensus among a panel of experts
    - The facilitator selects a group of experts based on the topic being examined
    - Each expert is sent a questionnaire to comment on the topic
    - Anonymous responses are returned to the facilitator who groups the comments and prepares copies of the information
    - The questionnaire and a copy of the compiled comments are sent back to each expert, along with the opportunity to comment further
    - At the end of each round, the facilitator decides if another round is necessary or if the results are ready

https://www.investopedia.com/terms/d/delphi-method.asp

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Tools and Techniques for Identifying Risks (3 of 3)

- Interviewing
  - Fact-finding technique for collecting information in face-to-face, phone, e-mail, or virtual discussions
  - Interviewing people with similar project experience is an important tool for identifying potential risks
- SWOT analysis
  - Strengths, weaknesses, opportunities, and threats
  - Helps identify the broad negative and positive risks that apply to a project

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Risk Register (1 of 5)

- Important output of the risk identification process
- Tool for documenting identified risk events and related information
  - Contains the results of various risk management processes
  - Is often displayed in a table or spreadsheet format
- Risk events refer to specific, uncertain events that may occur to the detriment or enhancement of the project

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Risk Register (2 of 5)

- Examples of negative risk events
  - Performance failure of a product
  - Delays in completing work as scheduled
  - Increases in estimated costs
  - Supply shortages
  - Litigation against the company
  - Strikes
- Examples of positive risk events
  - Completing work sooner or cheaper than planned
  - Collaborating with suppliers to produce better products
  - Good publicity resulting from the project

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Risk Register (3 of 5)

- Risk register contents for managing the project
  - Identification number for each risk event
  - Rank for each risk event
  - Name of each risk event
    - E.g. defective server
  - Description of each risk event
  - Category under which each risk event falls
    - E.g. Technology risk (hardware risk)
  - Root cause of each risk
    - E.g. defective power supply
  - Triggers for each risk
    - Triggers are indicators or symptoms of actual risk events
    - E.g. Users cannot get to the system
  - Potential responses to each risk
    - E.g. include a clause in the supplier's contract to replace the server within a certain time period at a negotiated cost
  - Risk owner or person who will own or take responsibility for each risk
  - Probability and impact of each risk occurring
  - Status of each risk

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Risk Register (4 of 5)

No.	Rank	Risk	Description	Category	Root Cause	Triggers	Potential Responses	Risk Owner	Probability	Impact	Status
R44	1										
R21	2										
R7	3										

Table 11-4 Sample risk register

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Risk Register (5 of 5)

- *No.*: R44
- *Rank*: 1
- *Risk*: New customer
- *Description*: We have never done a project for this organization before and don't know too much about them. One of our company's strengths is building good customer relationships, which often leads to further projects with that customer. We might have trouble working with this customer because they are new to us.
- *Category*: People risk
- *Root cause*: We won a contract to work on a project without really getting to know the customer.
- *Triggers*: The new customer asked a lot of questions in person and via e-mail that our existing customers would not, so we could easily misunderstand their needs and expectations.
- *Potential responses*: Make sure the project manager is sensitive to the fact that this is a new customer and takes the time to understand them. Have the PM set up a meeting to get to know the customer and clarify their expectations. Have Cliff attend the meeting, too.
- *Risk owner*: Project manager
- *Probability*: Medium
- *Impact*: High
- *Status*: PM will set up the meeting within the week.

Risk Report

- Risk report contents for communicating with selected audience, e.g. senior management
  - Sources of overall project risk
    - E.g. Unclear requirements or changing scope, Dependence on external vendors or new technology, Limited budget or schedule flexibility
  - Important drivers of overall project risk exposure
    - Key factors that amplify or reduce the total amount of risk, e.g. Team experience, Technology maturity, Stakeholder alignment, Dependency on third-party APIs
  - Summary information on risk events
    - Number of risks, e.g. 25 active risks
    - Total risk exposure, i.e. ∑(Probability×Impact)
    - Distribution across risk categories, e.g. 40% technical, 30% schedule, 20% cost, 10% external.
    - Metrics, e.g. % of risks mitigated, Risk exposure per sprint, Ratio of realized vs. identified risks
    - Trend, i.e. whether risk exposure is increasing or decreasing over time

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3. Performing Qualitative Risk Analysis

- Assess the likelihood and impact of identified risks to determine their magnitude and priority
- Risk quantification tools and techniques
  - Probability/impact matrix
  - Top Ten Risk Item Tracking
  - Expert judgment

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Probability/Impact Matrix or Chart (1 of 5)

- Lists relative probability of a risk occurring on one side of a matrix or axis on a chart and the relative impact of the risk occurring on the other
  - List the risks and then label each one as high, medium, or low in terms of its probability of occurrence and its impact if it did occur
  - May create a separate probability/impact matrix for negative risks and positive risks
- Process
  - Project stakeholders list the risks they think might occur
  - Then, they label each risk as having high, medium, or low probability and impact
  - Project manager summarizes the result
    - The team combine common risks and decide where those risks should be on the matrix
  - The team should focus on the risks that fall in the high sections
  - The team should discuss how it plans to respond to the risks if they occur
- Some project teams determine numeric scores of probability and impact of a risk to calculate Risk Index

$$risk\ index = probability\ of\ risk\ occuring * impact\ of\ risk\ occuring$$

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Probability/Impact Matrix or Chart (2 of 5)

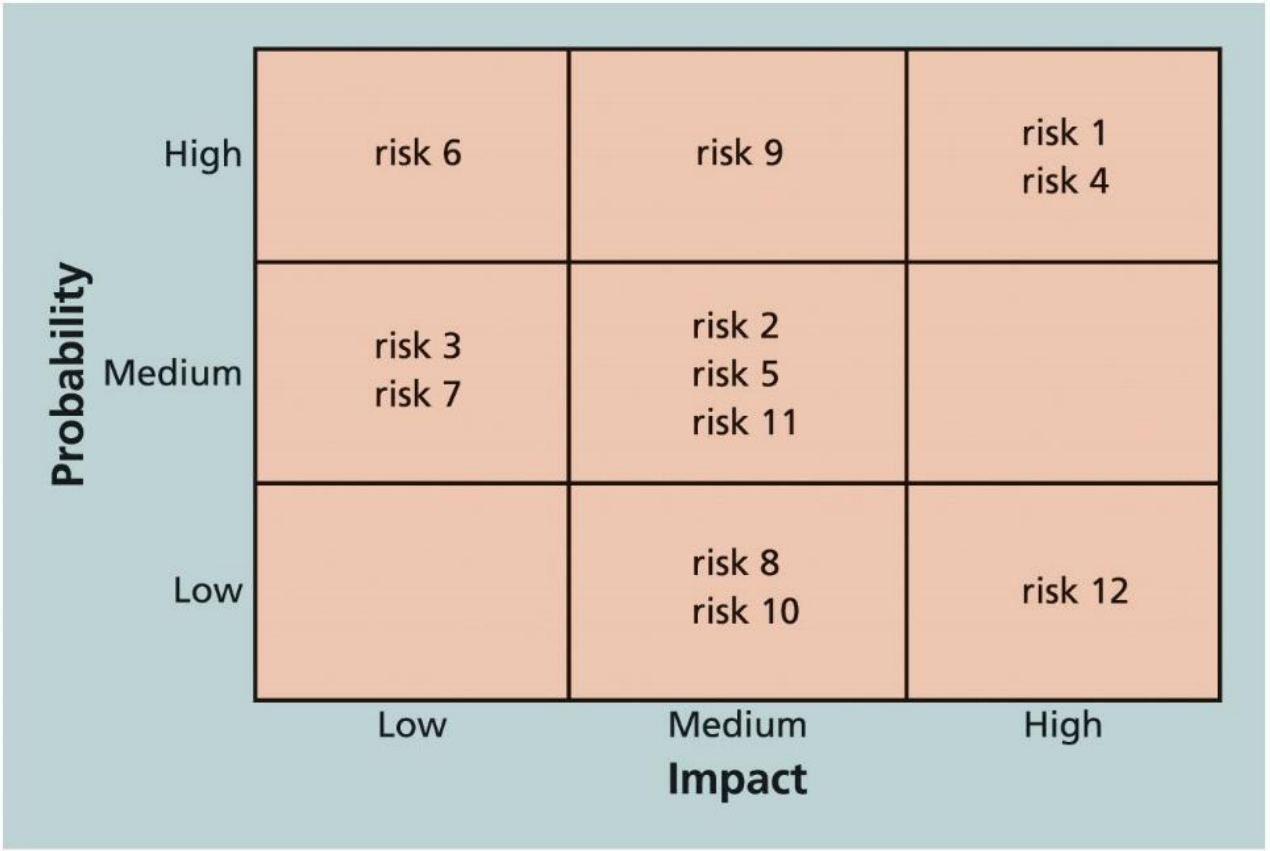


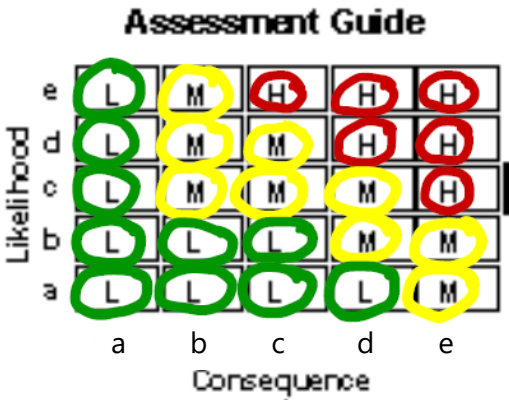
FIGURE 11-5 Sample probability/impact matrix

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Another method by DSMC Risk Management Guide for DoD Acquisition (Fourth Edition), February 2001
For determining the overall qualitative risk level of a negative risk

Level	What is the Likelihood the Risk Event Will Happen?
a	Remote
b	Unlikely
c	Likely
d	Highly Likely
e	Near Certainty

Process Variance refers to deviation from best practices. Likelihood/Probability refers to risk events.



Risk Assessment

R High - Unacceptable, Major disruption likely. Different approach required. Priority management attention required.

Y Moderate - Some disruption. Different approach may be required. Additional management attention may be needed.

G Low - Minimum impact. Minimum oversight needed to ensure risk remains low.

Level	Technical Performance	and/or	Schedule	and or	Cost	and or	Impact on Other Teams
a	Minimal or no impact		Minimal or no impact		Minimal or no impact		None
b	Acceptable with some reduction in margin		Additional resources required; able to meet need dates		<5%		Some impact
c	Acceptable with significant reduction in margin		Minor slip in key milestones; not able to meet need date		5-7%		Moderate impact
d	Acceptable, no remaining margin		Major slip in key milestone or critical path impacted		7-10%		Major impact
e	Unacceptable		Can't achieve key team or major program milestone		>10%		Unacceptable

Probability/Impact Matrix or Chart (4 of 5)

RISK MATRIX					
		SECTIONS IMPACTED			
		LOW	MEDIUM	HIGH	EXTREME
LIKELIHOOD	LOW	N	D	C	A
	MEDIUM	D	C	B	A
	HIGH	C	B	A	A

RISK MITIGATION BASED UPON GRADE	
GRADE	POSSIBLE ACTION
A	As a priority, mitigation actions reducing both likelihood and seriousness are to be identified and implemented at the start of project.
B	Mitigation actions reducing both likelihood and seriousness are to be identified and implemented throughout course of project.
C	Mitigation actions reducing both likelihood and seriousness are to be identified and costed for possible action should funds permit execution.
D	Risk to be noted with action only required should grading increase over time.
N	Risk to be noted with action only required should grading increase over time.

https://www.smartsheet.com/sites/default/files/2020-02/IC-MA-Project-Risk-Management-Plan-10750_PDF.pdf
https://www.smartsheet.com/all-risk-assessment-matrix-templates-you-need

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Probability/Impact Matrix or Chart (5 of 5)

1 – (Very unlikely): A very slim chance for this risk to occur.

2 – (Not likely): Low chances for this risk to occur.

3 – (Possible): Fifty-fifty chances for this risk to occur.

4 – (Probable): Good chances for this risk to occur.

5 – (Very likely): You can bet this risk will occur at some point.

1 – (Negligible): This risk will hardly impact your project.

2 – (Low): You can easily handle the consequences of this risk.

3 – (Moderate): It will take some time and effort to mitigate the consequences of this risk.

4 – (Significant): This risk could cause long-term consequences that will be hard to recover from.

5 – (Catastrophic): The impact of this risk might wreck your project.



1 – 6 (Low): Low-rating risks most likely will not happen. If they do, they will not be a threat to your project.

7 – 12 (Medium): Some medium-rating risks might happen at some point. You do not need to prioritize them but you should not ignore them either.

13 – 25 (High): High-rating risks are serious and very likely to happen threats. They can cause your project to go off the rails, so you should keep them in mind when planning your project.

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https://bigpicture.one/blog/project-risk-assessment-examples/

Top Ten Risk Item Tracking (1 of 2)

- Qualitative risk analysis tool that helps to identify risks and maintain an awareness of risks throughout the life of a project
  - Involves establishing a periodic review of the top ten project risk items
    - Current ranking
    - Previous ranking
    - Number of times the risk appears on the list over a period of time
    - Summary of progress made in resolving the risk item
- Qualitative analysis can also identify risks that should be evaluated quantitatively
- Low-priority risks that are still identified as potential risks are kept in a watch list

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Top Ten Risk Item Tracking (2 of 2)

	Monthly Ranking	Monthly Ranking	Monthly Ranking	
Risk Event	Rank This Month	Rank Last Month	Number of Months in Top Ten	Risk Resolution Progress
Inadequate planning	1	2	4	Working on revising the entire project management plan
Poor definition	2	3	3	Holding meetings with project customer and sponsor to clarify scope
Absence of leadership	3	1	2	Assigned a new project manager to lead the project after the previous one quit
Poor cost estimates	4	4	3	Revising cost estimates
Poor time estimates	5	5	3	Revising schedule estimates

Table 11-5 Example of top ten risk item tracking (top five negative risks)

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4. Performing Quantitative Risk Analysis

- Often follows qualitative risk analysis, but both can be done together
  - Large, complex projects involving leading edge technologies often require extensive quantitative risk analysis
- Main techniques
  - Decision tree analysis
  - Simulation
  - Sensitivity analysis

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Decision Trees and Expected Monetary Value (EMV) (1 of 2)

- A decision tree is a diagramming analysis technique used to help select the best course of action in situations in which future outcomes are uncertain
- Estimated monetary value (EMV) is the product of a risk event probability and the risk event's monetary value
  - You can draw a decision tree to help find the EMV

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Decision Trees and Expected Monetary Value (EMV) (2 of 2)

Decision tree helps to select the best choice, i.e. highest EMV

Risk analysis at project bidding selection

Time and resources are needed to respond to request for proposals (RFPs) from prospective clients

EMV is calculated from probability of winning/losing bid and consequence as profit/loss

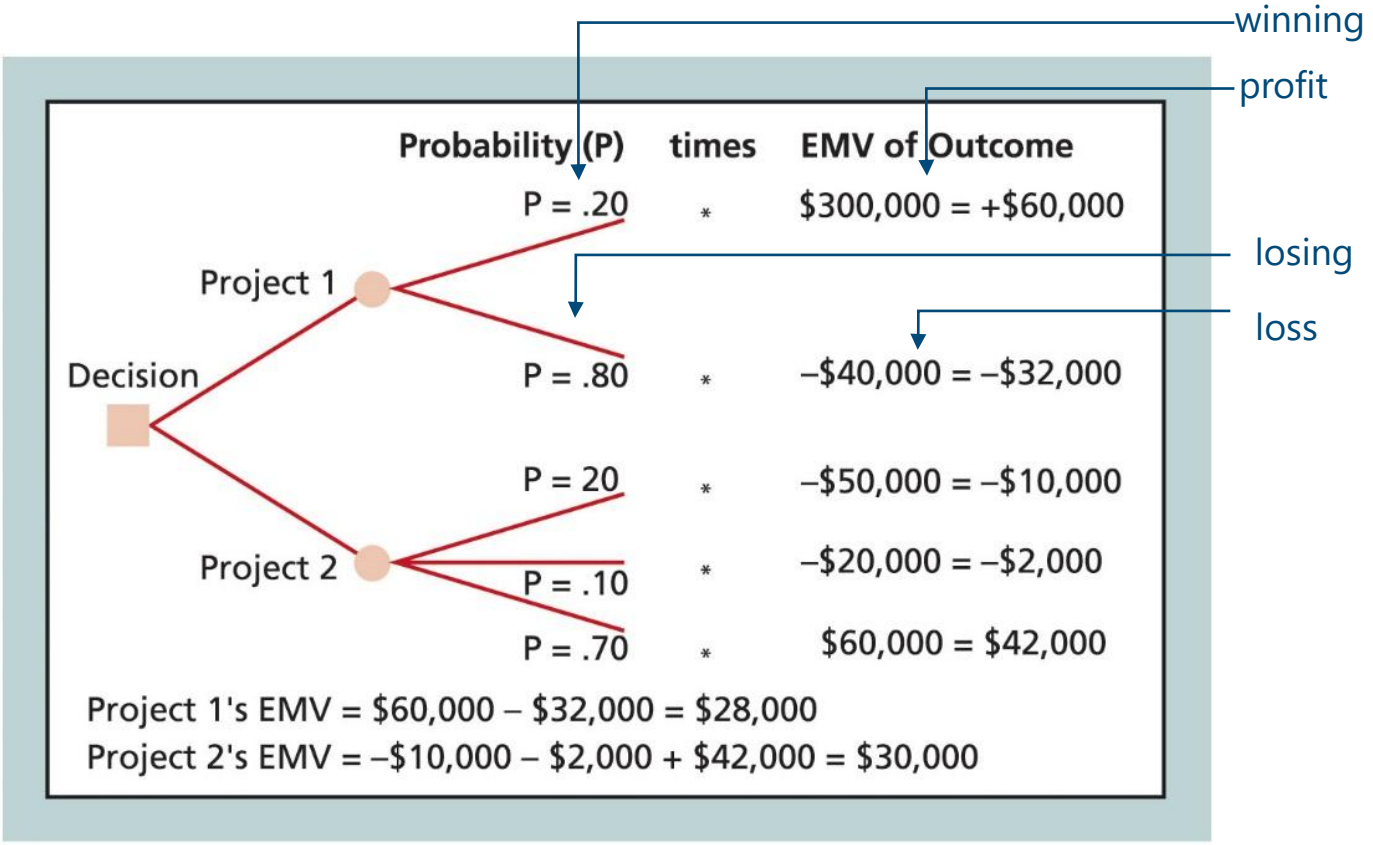


FIGURE 11-7 Expected monetary value (EMV) example

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Simulation (1 of 5)

- Uses a representation or model of a system to analyze the expected behavior or performance of the system
  - Monte Carlo analysis simulates a model's outcome many times to provide a statistical distribution of the calculated results
    - Predict the probability of finishing the project by a certain date or the probability that the cost will be equal to or less than a certain value
    - You can use several different types of distribution functions when performing a Monte Carlo analysis
    - Microsoft Excel is a common tool for performing quantitative risk analysis

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Simulation (2 of 5)

- Steps of a Monte Carlo analysis
  1. Identify a problem for risk analysis and variables in the model (such as time and cost)
     - E.g. the problem is the likelihood of meeting project schedule goal (and the model is the project network diagram and the variables are the time of tasks)
  2. Collect the most likely, optimistic, and pessimistic estimates for the variables in the model (such as time or cost of each task)
     - E.g. an expert might assign to a task the optimistic estimate of 8 weeks, the most likely estimate of 10 weeks, and the pessimistic estimate of 15 weeks
  3. Determine the probability distribution of each variable
     - E.g. the expert might determine the probability of completing this task in [8, 10) weeks is 20%, and in [10, 15] weeks 80%
  4. Select a random value for each variable based on the probability distribution for each variable
     - E.g. for the time estimate of this task, randomly pick a value in [8, 10) weeks 20% of the time and in [10, 15] weeks 80% of the time
  5. Run a deterministic analysis or one pass through the model using the values selected for all variables
     - E.g. this task might have the value of 12 on the first run (for calculating the total project time)
  6. Repeat steps 4-5 many times (typically between 100 and 1,000) to obtain the probability distribution of the model's results

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Simulation (3 of 5)

Duration (days)



Optimistic	8	20%		5	30%		60%
Most likely	10			9		8	
Pessimistic	15	80%		12	70%	12	40%

Round 1	12	20 rounds [8, 10) 80 rounds [10, 15]	10	10	32
…	…		…	…	…
Round 100	9		…	…	…

Project durations are used to plot bar chart of sample counts, percentage, or cumulative percentage (probability)

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Simulation (4 of 5)

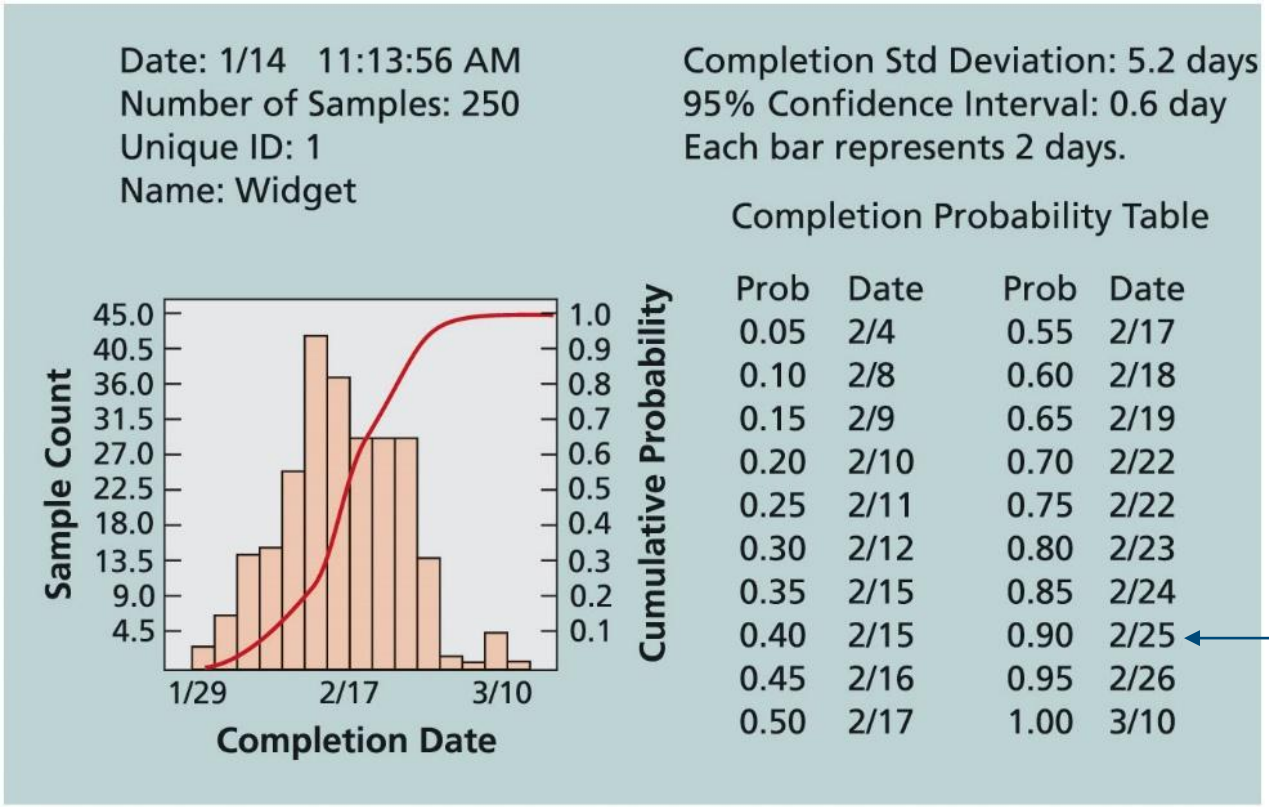


FIGURE 11-8 Sample Monte Carlo–based simulation results for project schedule

90% probability that the project will finish by 2/25

(Project manager should select 90% chance of finish)

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Simulation (5 of 5)

Here, each task cannot start before the preceding one is complete, so the estimated duration for completion of all three tasks is 17 weeks. However, if we now consider a range of uncertainty in duration for completing each task, the schedule will look like this:

Activity	Duration (Weeks)	Early Finish (Weeks)	Late Finish (Weeks)
Task #1	5	4	6
Task #2	8	5	10
Task #3	4	3	6
Total	**17**	**12**	**22**

Most likely

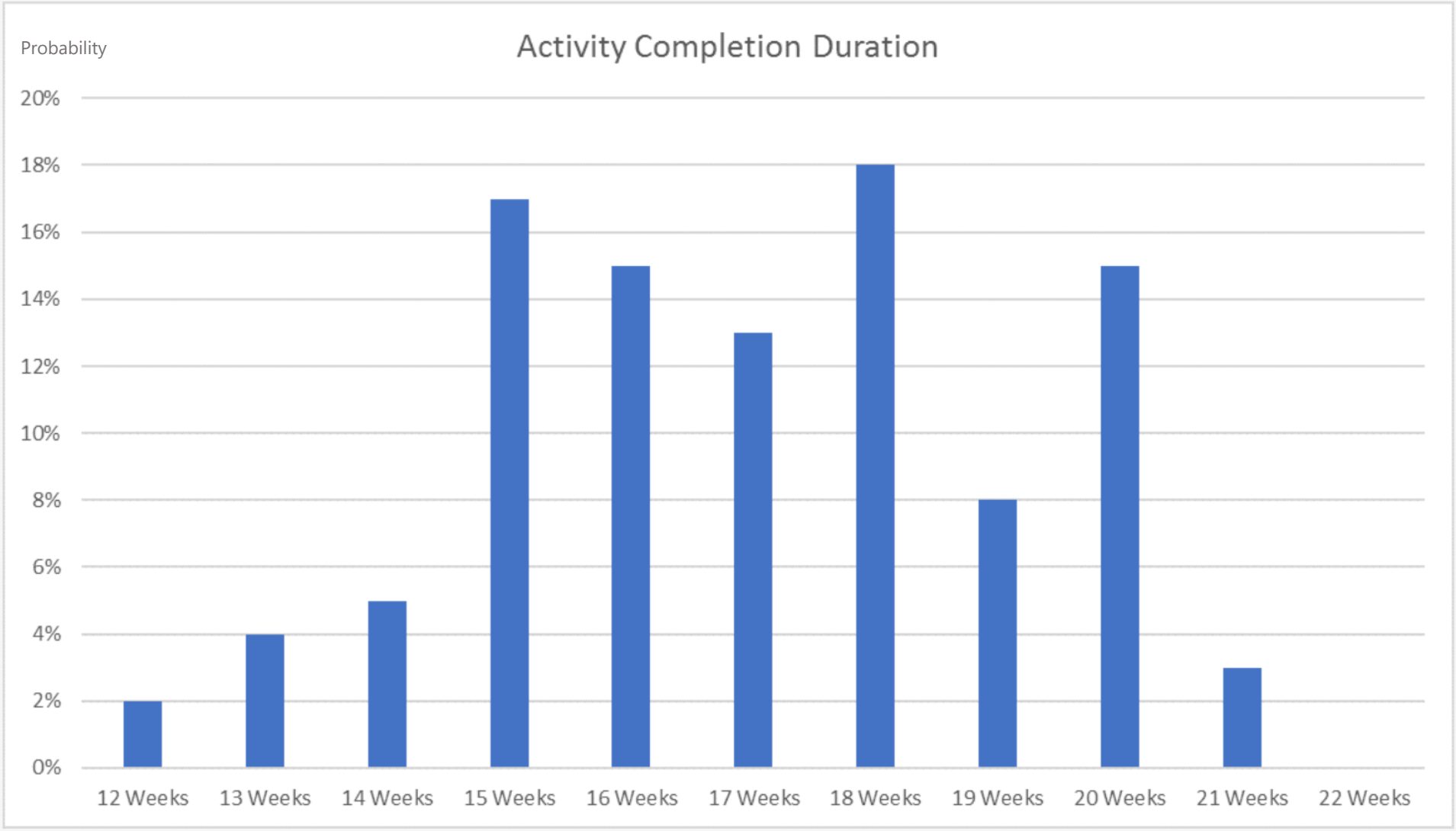
Optimistic

Pessimistic

https://www.project-risk-manager.com/blog/monte-carlo-simulation/

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Running a Monte Carlo simulation for each activity 100,000 times across the range of duration uncertainty, we end up with a distribution of the outcomes which shows us that the most likely duration for all three tasks to be completed is 18 weeks, and not 17 as was initially estimated.



Monte Carlo Simulation - Most Likely Completion Time

https://www.project-risk-manager.com/blog/monte-carlo-simulation/

Another way of looking at this, would be to consider the cumulative probability distribution curve. This gives us an indication of the probability of completing the activity within a certain time-frame.



Monte Carlo Cumulative Probability Distribution

From the cumulative probability curve shown above, we can establish that there is a 50% chance of completing the activity within 17 weeks, and a 90% chance of completing the activity within 20 weeks.

https://www.project-risk-manager.com/blog/monte-carlo-simulation/

Sensitivity Analysis (1 of 2)

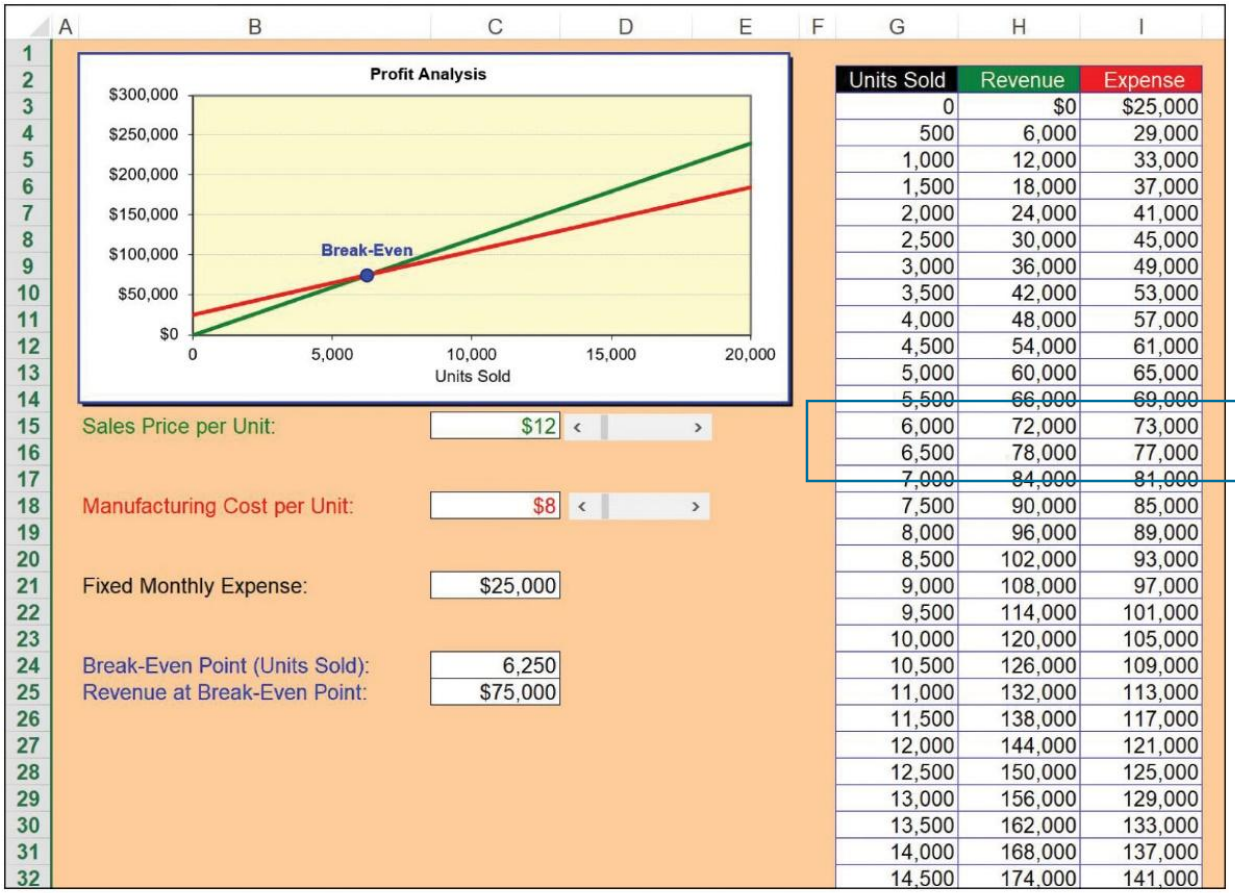
- Used to show the effects of changing one or more variables on an outcome
  - To help make common business decisions, such as break-even points based on different assumptions
    - E.g. determining profit by varying number of products produced, number of hours required to do the jobs, costs per hour
  - Spreadsheet software, such as Microsoft Excel, is a common tool for performing sensitivity analysis

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Sensitivity Analysis (2 of 2)

Determining opportunity to make profit in each month by varying units sold

(>6250 units)



Units Sold	Revenue	Expense
0	$0	$25,000
500	6,000	29,000
1,000	12,000	33,000
1,500	18,000	37,000
2,000	24,000	41,000
2,500	30,000	45,000
3,000	36,000	49,000
3,500	42,000	53,000
4,000	48,000	57,000
4,500	54,000	61,000
5,000	60,000	65,000
5,500	66,000	69,000
6,000	72,000	73,000
6,500	78,000	77,000
7,000	84,000	81,000
7,500	90,000	85,000
8,000	96,000	89,000
8,500	102,000	93,000
9,000	108,000	97,000
9,500	114,000	101,000
10,000	120,000	105,000
10,500	126,000	109,000
11,000	132,000	113,000
11,500	138,000	117,000
12,000	144,000	121,000
12,500	150,000	125,000
13,000	156,000	129,000
13,500	162,000	133,000
14,000	168,000	137,000
14,500	174,000	141,000

FIGURE 11-9 Sample sensitivity analysis for determining break-even point

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5. Planning Risk Responses (1 of 5)

- After identifying and quantifying risks, the organization must decide how to respond to them
  - Basic response strategies for negative risks
    - Risk avoidance
      - Eliminating cause of a risk
      - E.g. continue using familiar pieces of hardware and software instead of unfamiliar ones
    - Risk acceptance
      - Accepting consequences if a risk occurs
      - E.g. use a backup location if the project review meeting site is not available
    - Risk transference
      - Shifting risk consequence and management responsibility to a third party
      - E.g. purchase special insurance or warranty protection for specific hardware
    - Risk mitigation
      - Reducing risk impact by reducing probability of occurrence
      - E.g. use proven technology, use competent project personnel, use various analysis and validation techniques, buy maintenance or service agreements from subcontractors
    - Risk escalation
      - Notifying higher level of authority if the risk is outside the project scope, or the risk response is outside the project manager's authority

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Planning Risk Responses (2 of 5)

Technical Risks	Cost Risks	Schedule Risks
Emphasize team support and avoid stand-alone project structure	Increase the frequency of project monitoring	Increase the frequency of project monitoring
Increase project manager authority	Use WBS and CPM	Use WBS and CPM
Improve problem handling and communication	Improve communication, understanding of project goals, and team support	Select the most experienced project manager
Increase the frequency of project monitoring	Increase project manager authority	
Use WBS and CPM		

Table 11-6 General risk mitigation strategies for technical, cost, and schedule risks. *Source: J. Couillard

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Planning Risk Responses (3 of 5)

- Basic response strategies for positive risks
  - Risk exploitation
    - Eliminate the uncertainty associated with the risk to ensure it occurs
    - E.g. organize a news coverage and public event of the project that provides new computer classrooms for a school in need (for good publicity that could lead to more business)
  - Risk sharing
    - Allocating ownership of the risk to another party
    - E.g. form partnership with the school's board and parent-teacher organization to share responsibility for good publicity
  - Risk enhancement
    - Increase the probability or the positive impacts of an opportunity
    - E.g. generate awareness and excitement among students, parents, teachers who might do their own advertising of the project
  - Risk acceptance
    - Not taking any actions toward the risk
    - E.g. assume the project will result in good publicity and not do anything extra
  - Risk escalation
    - Transfer risks from the project level to higher organizational levels
    - E.g. escalate risk of partnering with another firm (e.g. firm that provides learning content) to a higher-level manager who has more influence with the firm

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Planning Risk Responses (4 of 5)

- It is also important to identify residual risks and secondary risks
  - Residual risks: risks that remain after all of the response strategies have been implemented (the response strategies fail or are not fully effective)
    - E.g. even though a stable hardware product is used (risk mitigation), there may still be a risk that the system fails to function properly
  - Secondary risks: risks that are created directly by implementing a risk response
    - E.g. using the more stable hardware may have caused a risk of peripheral devices failing to function properly

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Planning Risk Responses (5 of 5)

- Additional plans
  - Contingency plans
    - Predefined actions that the project team will take if an identified risk event occurs
    - E.g. use a third party backup service to save data for the web site in case the hosting provider does not have any backup
  - Fallback plans
    - Predefined actions that are put into effect if attempts to reduce the risk (i.e. contingency plans) fail or are not effective
    - E.g. use the copy on the computer or google drive in case the backup service has gone out of business
  - Contingency reserves or allowances (or known unknowns)
    - Funds included in the cost baseline that can be used to mitigate cost or schedule overruns if identified risks occur
    - An estimate reserve, not random, based on various risk management techniques, e.g. expected monetary value (EMV)
  - Management reserves (or unknown unknowns)
    - Funds held for unidentified risks that are used for management control purposes, e.g. 5-10% of total project cost

https://pmstudycircle.com/2012/02/contingency-plan-vs-fallback-plan/
https://pmstudycircle.com/2012/02/contingency-reserve-vs-management-reserve/

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6. Implementing Risk Responses

- Main executing process performed as part of project risk management is implementing risk responses
  - Key outputs: change requests, project documents updates
  - E.g. for the new customer risk, perform risk mitigation
    - Make sure the project manager takes time to understand the customer
    - Set up a meeting to get to know the customer
    - Clarify expectation

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7. Monitoring Risks

- Project risk management does not stop with the initial risk analysis
- Monitoring risks based on defined milestones and making decisions regarding risks and their response strategies
  - Ensuring the appropriate risk responses are performed
  - Tracking identified risks
  - Identifying and analyzing new risk
  - Evaluating effectiveness of risk management throughout the entire project
    - Altering a strategy that becomes ineffective
    - Implementing a contingency plan and fallback plan
    - Eliminating a risk from the list of potential risks when it no longer exists
    - Using workarounds—unplanned responses to risk events—when they do not have contingency plans in place

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Using Software to Assist in Project Risk Management

- Risk registers can be created in a simple Microsoft Word or Excel file or as part of a sophisticated database
  - More sophisticated risk management software, such as Monte Carlo simulation tools, help develop models and use simulations to analyze and respond to various risks

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Considerations for Agile/Adaptive Environments

- Risk is considered when selecting content of each iteration
- Using frequent reviews of incremental work products and cross-functional teams to accelerate knowledge sharing and ensure that risk is understood and managed
- Risks are identified, analyzed, and managed during each iteration

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Chapter Summary

- Risk is an uncertainty that can have a negative or positive effect on meeting project objectives
  - Many organizations do a poor job of project risk management, if they do any at all
  - Successful organizations realize the value of good project risk management
- Risk management is an investment
  - Costs are associated with identifying risks, analyzing those risks, and establishing plans to address them
- Implementing risk responses involves putting the appropriate risk response plans into action
  - Monitoring risks involves monitoring implementation of risk response plans, tracking identified risks, identifying and analyzing new risks, and evaluating effectiveness of risk management throughout the entire project

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